MEETFLEET RESEARCH DIVISION TECHNICAL WHITE PAPER

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The Social Activation Score (SAS)

A Proprietary Multi-Dimensional Matching Function for Real-World Human Connection

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ABSTRACT

We introduce the Social Activation Score (SAS), a novel proprietary scoring function developed by MeetFleet to address the computational challenge of matching individuals for real-world shared activities. Unlike conventional recommendation systems that optimize for engagement or preference similarity alone, SAS jointly optimizes for psychological vibe alignment, intent-to-activity compatibility, logistical friction minimization, and agent-verified environmental safety. The result is a single, interpretable score that determines the real-world feasibility of two people doing something together at a specific place, at a specific time.

1. INTRODUCTION

The dominant paradigm in social matching filtering users by static attributes such as shared interests, demographics, or stated preferences has proven insufficient for facilitating genuine real-world connection. Platforms built on this model optimize for digital engagement rather than physical activation, leaving the core problem of loneliness structurally unsolved.

MeetFleet reframes the problem entirely. The unit of value is not a match between two people, but a *successful shared activity* a jog completed, a study session held, a market visited together. This shift from being to *doing* demands a fundamentally different scoring architecture, one that accounts for the full logistical and social complexity of real-world human coordination.

The Social Activation Score (SAS) is that architecture.

2. THE SAS MASTER FORMULA

DEFINITION 1 SOCIAL ACTIVATION SCORE

$$SAS(u_a,\, u_b,\, v) \;=\; \cos(\mathbf{U}_{ab},\, \mathbf{V}_v) \cdot\; \Phi(I_{ab},\, A_v) \cdot\; \Psi(f,\, \delta,\, \tau) \cdot\; \Lambda(\rho,\, \sigma) \tag{1}$$

Where u_a and u_b are the two matched users and v is the candidate venue returned by the physical agent audit. The score $SAS \in [0, 1]$, where values approaching 1 indicate maximum real-world activation potential.

3. COMPONENT DEFINITIONS

3.1 Vibe Alignment $\cos(\mathbf{U}_{ab}, \mathbf{V}_v)$

The first term applies Cosine Similarity between the merged user preference vector and the venue's high-dimensional vibe vector, as captured by the MeetFleet physical agent during venue audit:

$$\cos(\mathbf{U}_{ab}, \mathbf{V}_v) \;=\; \frac{\mathbf{U}_{ab} \cdot \mathbf{V}_v}{\|\mathbf{U}_{ab}\|\;\|\mathbf{V}_v\|} \tag{2}$$

$\mathbf{U}_{ab}$	The merged preference vector of User A and User B, representing their combined Vibe Signature.
$\mathbf{V}_v$	The venue's vibe vector constructed from physical agent field data (ambiance, noise, lighting, crowd profile).
$\|\cdot\|$	The Euclidean norm, used to normalize both vectors and ensure direction-only comparison.

3.2 Intent-to-Activity Fit $\Phi(I_{ab}, A_v)$

The second term penalizes mismatches between the users' shared intent and the venue's verified activity compatibility profile. A venue suitable for loud socializing should score low

for a quiet study session, regardless of vibe alignment.

$$\Phi(I_{ab}, A_v) = 1 - \frac{\|I_{ab} - A_v\|}{n} \tag{3}$$

I_{ab}	The shared intent vector of User A and B (e.g., jogging, grocery run, quiet work session).
A_v	The activity compatibility vector of the venue, verified by the physical agent on-site.
n	The number of intent dimensions, used to normalize the penalty to $[0, 1]$.

3.3 Friction Coefficient $\Psi(f, \delta, \tau)$

The third term models the total logistical and financial friction of the proposed meetup using a decaying exponential. High friction financial imbalance, geographical unfairness, or scheduling misalignment exponentially dampens the final score.

$$\Psi(f, \delta, \tau) = e^{-(\alpha f + \beta\delta + \gamma\tau)} \tag{4}$$

f	Financial friction score. Zero if the universal payment contract selects a free or split model; increases with financial asymmetry.
δ	Distance delta the geographical unfairness of the venue relative to both users' locations.
τ	Time misalignment between the available scheduling windows of User A and User B.
α, β, γ	Tunable weighting constants controlled by MeetFleet R&D, adjustable per activity type or market context.

3.4 Safety & Social Optimization Layer $\Lambda(\rho, \sigma)$

The fourth term is MeetFleet's most structurally novel contribution. It evaluates the agent-verified environmental safety and social density of the venue using a self-regulating function that peaks at balanced conditions and degrades at both extremes.

$$\Lambda(\rho, \sigma) = \frac{\rho \cdot \sigma}{1 + |1 - \rho \cdot \sigma|} \tag{5}$$

ρ	Safety score of the venue as assessed by the physical agent (lighting, accessibility, crowd behavior, emergency access).
σ	Social density score calibrated for the target activity. Neither too empty nor too crowded.

4. COMPONENT SUMMARY

COMPONENT	NAME	WHAT IT MEASURES
`cos(U, V)`	Vibe Alignment	Psychological and aesthetic fit between users and venue
`Φ(I, A)`	Intent-to-Activity Fit	Whether the venue actually supports the planned activity
`Ψ(f, δ, τ)`	Friction Coefficient	Financial, geographical, and temporal cost of the meetup
`Λ(ρ, σ)`	Safety & Social Layer	Agent-verified environmental safety and crowd calibration

5. MATHEMATICAL PROPERTIES

FORMAL PROPERTIES OF SAS

→ **Boundedness:** $SAS \in [0, 1]$ for all valid inputs, enabling direct score comparison across user pairs and venues.

→ **Multiplicative structure:** Any single component approaching zero collapses the full score an unsafe venue or extreme friction cannot be compensated by strong vibe alignment alone.

→ **Differentiability:** All four components are differentiable with respect to their inputs, enabling gradient-based optimization and continuous learning from outcome data.

→ **Modularity:** Each component can be independently weighted, retrained, or swapped, allowing the R&D team to iterate on sub-functions without rebuilding the full pipeline.

6. CONCLUSION

The Social Activation Score represents the first mathematically formalized, agent-grounded scoring function designed explicitly for real-world social logistics. By jointly modeling psychological alignment, behavioral intent, logistical friction, and physical environment quality, SAS operationalizes MeetFleet's core thesis: that meaningful human connection is an engineering problem, and one that is now solvable at scale.

"SAS is a multi-dimensional activation function that scores the real-world feasibility of two people doing something together, accounting for psychological alignment, intent fit, logistical friction, and agent-verified environmental safety simultaneously."